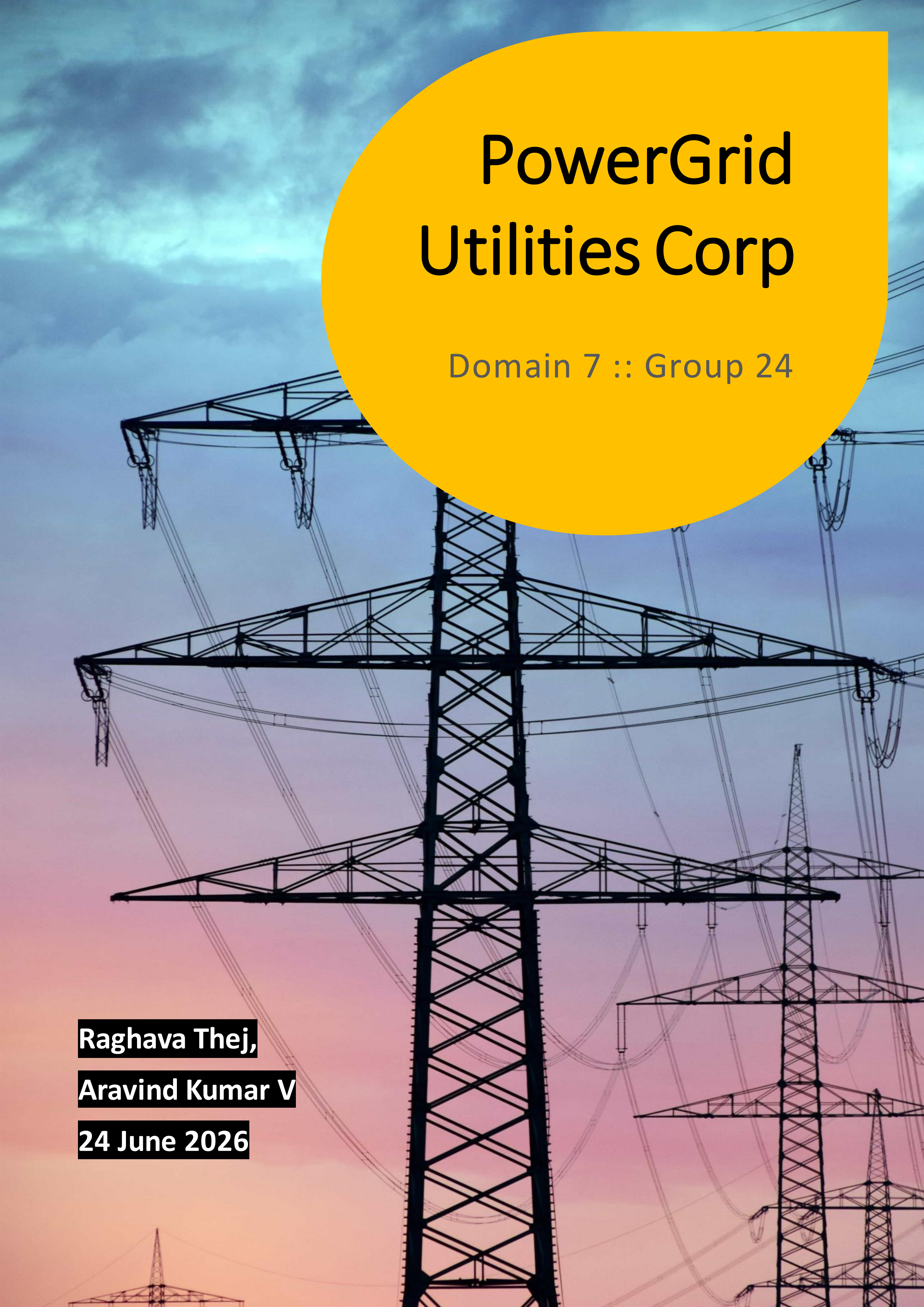

PowerGrid Utilities Corp

Domain 7 :: Group 24

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24 June 2026

Capstone Project : Pre-submission Plan

01 : Business Problem Understanding & Objective Formulation

Powergrid Utilities Corporation is facing variety of problems while serving its customers. The situation has reached this level due to many reasons (Cause) and resulting in many day-to-day & long-term problems (Effect)

Cause

- ✓ Increasing Urbanization
- ✓ Industrial Growth
- ✓ Renewable Energy Integration
- ✓ Climate Disruptions

Effect

- Voltage Instabilities & Equipment Failures
- Service Interruptions
- Regulatory Issues & Customer Complaints
- Weather Risks

<u>Business Problems</u> (based on dataset columns)

- ✓ Frequent Asset Failures result service interruptions (Average outage & Previous outage)
- ✓ Outages impact on revenue and customers (Customers served & Estimated revenue loss)
- ✓ Aging / Degrading assets not managed proactively (Asset age, Equipment & Oil quality score)
- ✓ Assets are operated under high stress conditions (Power Load & Load Utilization)

<u>Business Objective</u>

- Reduce high impact grid failures & outages
- Improve Asset Reliability and work on increasing it
- Optimize Maintenance & Operational Decision making

<u>Analytical Objective</u>

- Identify Key drivers of grid failures
- Building a predictive Asset Level Failure Risk Model
- Generate Risk scores and prioritize assets based on business impact

02 : Challenges & Constraints

- Date range of the sourced data is not present
- Realtime data would be more useful as ongoing anomalies can be spotted
- Missing values of many columns such as oil quality score, inspection score
- Failure data behaviour is complex across mixed asset types data
- Integration of predictions/findings into existing system would be challenging
- Maintenance Teams must culturally shift from reactive to predictive approach
- Tuning the model from false positives and false negatives

03 : Background Study / Literature Review

Predictive maintenance has become an important strategy in the utility industry for reducing equipment failures and improving grid reliability. By analysing operational, maintenance, and environmental data, utilities can identify high-risk assets before failures occur and schedule maintenance proactively.

Several machine learning techniques have been widely applied in predictive maintenance:

Logistic Regression (LR) A simple and interpretable classification technique commonly used as a baseline model. However, it may struggle to capture complex non-linear relationships in operational data.	**Decision Trees (DT)** Decision Trees provide easy-to-understand rules and explanations but may suffer from instability and overfitting when dealing with complex datasets.
Support Vector Machines (SVM) A powerful classification method capable of handling non-linear patterns through kernel functions. Although often accurate, it can be computationally expensive and sensitive to class imbalance.	**Random Forest (RF)** Combining multiple DT to improve prediction accuracy and reduce overfitting. It performs well with noisy & heterogeneous utility data and is widely used in predictive maintenance applications.

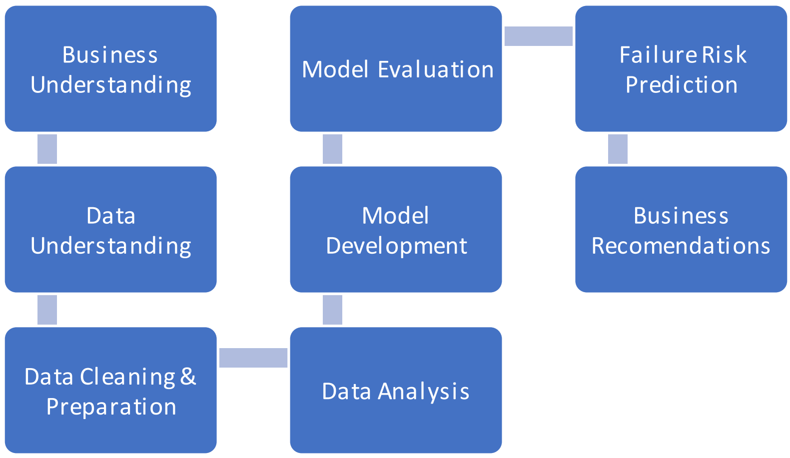
04 : Proposed Methodology / Solution Approach

<u>Problem Framing :</u>

- Primary Modelling : We are going to do Binary classification on grid_failure_flag's using asset, operational, maintenance & weather variables.

- Secondary Outputs : Calculate Risk scores by asset type and substation region and calculate estimated impact for high-risk assets based on revenue loss, penalty cost & customer served.

<u>Process workflow :</u>



<u>Expected Outcome</u>

The solution will identify assets with a high probability of failure, helping maintenance teams prioritize inspections and repairs, reduce outages, and improve overall grid reliability

05 : Proposed Timeline

- ✓ <u>Week 1–2:</u> Data understanding & cleaning (EDA, handling missing values, initial visualizations).
- ✓ <u>Week 3–4</u>: Feature engineering & baseline models (logistic regression, decision tree).
- ✓ <u>Week 5–6:</u> Model Development, evaluation, and comparative analysis (e.g., tree-based models, importance).
- ✓ <u>Week 7:</u> Business interpretation, risk scoring, and prioritization framework.
- ✓ <u>Week 8:</u> Documentation & Final Presentation

06 : Expected Deliverables / Outputs

<u>Final Deliverables :</u>

- ✓ Final project report
- ✓ Technical documentation (Feature Engineering)
- ✓ Business recommendation report (Grid failure, High Risk Asset & Key Drivers)
- ✓ Cleaned Dataset, EDA Report, ML Models (ipynb) & Model Evaluation Report
- ✓ Final presentation